Rank-seven Grothendieck rings: certified exclusions and models through multiplicity four

Abstract

We apply general categorification obstructions to the pinned, complete fusion-ring census of rank seven and multiplicity at most four. Every asserted exclusion has an exact certificate or a cited theorem with exactly checked tensor hypotheses; every asserted realization has an explicit unitary fusion-category model and an exact full-tensor identification. We include a rank-seven extension of the formal-codegree test to noncommutative rings, a global-dimension-fifteen obstruction, and complete oriented signed-sphericalization certificates for 246 rings through multiplicity four. The remaining tensors are recorded as *undecided*. Three additional published AnyonWiki associators have independent exact pentagon, invertibility, unit and rigidity checks. Thus multiplicity one is complete, with thirteen realized rings; the first undecided multiplicity is two.

1 Scope and statements

Let R have distinguished basis $(b_0 = 1, b_1, \dots, b_6)$ and multiplication $b_i b_j = \sum_k N_{ij}^k b_k$. A *Grothendieck ring* here means the based ring of a complex fusion category; no pivotal, spherical, braided, or unitary hypothesis is placed on a hypothetical categorification. Its multiplicity is $\max_{i,j,k} N_{ij}^k$. The upstream *Fusion Ring Census* [14] provides the complete rank-seven fusion-ring list through multiplicity four. We pin its source bytes, source commit, row ordering, and their digest in the companion package. This is a categorification study of that census; the list of undecided tensors is not counted as a list of Grothendieck rings. Multiplicities at least five lie outside the pinned source and the claims of this manuscript.

Theorem 1 (Certified rank-seven status through multiplicity four)**.** *The numbers of excluded, explicitly unitarily realized, and undecided rank-seven fusion rings are given by Table 1. The negative conclusions hold for* every *complex fusion category. The twenty-six positive rows have the models in Table 3, matched against their complete fusion tensors. No conclusion is drawn for the undecided rows.*

Table 1: Certified status of complete rank-seven fusion-ring census through exact multiplicity four. Undecided entries are not assumed categorifiable.

Exact multiplicity	Fusion rings	Excluded	Realized	Undecided
1	43	30	13	0
2	319	295	9	15
3	1059	1039	2	18
4	2780	2752	2	26

Table 2: Sequential exclusion counts, charged only to the first successful criterion. The final six rows record the further theoretical update.

Criterion	$m=1$	$m=2$	$m=3$	$m=4$
Near-group theorem	2	1	1	1
Global dimension 15	1	0	0	0
Zero spectrum	8	30	51	63
Closed rank-four/five subring	0	25	125	235
Formal-codegree d-number	15	179	788	2314
Subcategory dimension quotient	0	5	9	8
Oriented dimension order	4	51	61	130
Localized triangular-prism equations	0	1	1	0
Solvability and support inequality	0	0	2	0
Half-partition Tannakian criterion	0	0	1	0
Odd pointed group in dimension 120	0	0	0	1
Double equivariantization in dimension 52	0	1	0	0
Order-sixteen orbit obstruction	0	2	0	0

Table 3: Unitary model constructions identified by exact full-tensor based-ring isomorphism. Each row denotes a ring, not a count of categories.

Census ID	Explicit model
`R7_1_1`	$\mathrm{Rep}(D_{16})$
`R7_1_6`	$\mathrm{SU}(2)_6$
`R7_1_9`	$\mathrm{PSU}(2)_{12}$
`R7_1_12`	$\mathrm{SO}(7)_2$
`R7_1_16`	$\mathrm{Rep}(D_{22})$
`R7_1_17`	$\mathrm{PSU}(2)_{13}$
`R7_1_19`	AnyonWiki $\mathrm{FC}^{7,1,2}_{3,1,0,1}$
`R7_1_24`	$\mathrm{Zest}(\mathrm{SU}(2)_6)$
`R7_1_25`	$\mathrm{Rep}(SD_{16})$
`R7_1_28`	AnyonWiki $\mathrm{FC}^{7,1,2}_{12,1,0,1}$
`R7_1_36`	AnyonWiki $\mathrm{FC}^{7,1,4}_{3,1,0,1}$
`R7_1_39`	$\mathrm{TY}(C_6)$
`R7_1_43`	$\mathrm{Vec}(C_7)$
`R7_2_1`	TY parity equivariantization (§5.1)
`R7_2_24`	$\mathrm{Rep}(S_5)$
`R7_2_195`	TY twisted-C_4 equivariantization (§5.1)
`R7_2_218`	$\mathrm{Rep}(C_{13} \rtimes C_4)$
`R7_2_233`	$\mathrm{Rep}(H_{24})$
`R7_2_234`	$\mathcal{C}(A_5, 1, A_4, 1)$
`R7_2_297`	$\mathrm{Rep}(SL_2(\mathbb{F}_3))$
`R7_2_313`	$\mathrm{Adj}(\mathrm{SU}(3)_5)$
`R7_2_319`	$\mathrm{Rep}(C_{13} \rtimes C_3)$
`R7_3_602`	$\mathrm{Rep}(A_6)$
`R7_3_1059`	$\mathrm{Rep}(C_{11} \rtimes C_5)$
`R7_4_2`	$\mathrm{Vec}(Q_8)^{S_3}$
`R7_4_2019`	Twisted quaternion model (§5.3)

Multiplicity one is complete: its former residuals $R7_1_19$, $R7_1_28$, and $R7_1_36$ have

the AnyonWiki models described below. The first undecided multiplicity is two. The exhaustive source-position convention is $R7_m_j$, where j is the position *among rows of exact multiplicity* m in the pinned census file. This convention makes the positive-model permutations and negative witnesses unambiguous.

2 Necessary conditions and their order

The checks in Table 2 run from top to bottom on surviving tensors, and a tensor is charged to its first successful condition. Near-group and small-dimension checks read a few coefficients; the zero-spectrum and closed-subring passes are finite tensor scans; exact Casimir factoring and subcategory-dimension arithmetic follow. The complete oriented search ends this initial sequence; the further theoretical exclusions are appended as a dated update. This order records precisely when a criterion first contributes rather than hiding it in an unordered union of tests. Multiplicity two is still undecided, so these stages do *not* establish a multiplicity threshold beyond which no new criteria would be needed. The oriented criterion first contributes at multiplicity one, so it cannot be deferred to a higher multiplicity without losing a proof.

If G is the group of invertible basis elements and the only other object X satisfies $X^2 = \sum_{g\in G} g + \ell X$, then for $|G| = 6$: at $\ell = 0$ the Tambara–Yamagami construction forces G abelian [27]; at $0 < \ell < 5$ in the commutative case, Schopieray's near-group obstruction rules out categorification [26, Proposition 3.2]. These statements apply to a categorification without a pivotal assumption. The certificate records the six group elements, the remaining object and ℓ, and rechecks every fusion-rule hypothesis.

The global-dimension-fifteen branch has six dimension-one simples and one dimension-three simple: direct substitution into every product establishes that $(1, \dots, 1, 3)$ is its strictly positive dimension character. Hence its Frobenius–Perron dimension is $6 + 3^2 = 15$. The *pq* classification [19, Theorems 3.1 and 5.1] makes a complex fusion category of dimension $3 \cdot 5$ pointed in this case, contradicting the dimension-three simple.

The zero-spectrum obstruction is the exact necessary criterion of Liu–Palcoux–Ren [22]; each certificate stores its nine basis indices and the verifier rechecks all required zero/nonzero fusion coefficients. A proper fusion-closed subset of basis elements would generate a fusion subcategory. If its induced rank-four or rank-five ring is excluded by the complete lower-rank classifications [15, 16], the ambient ring is excluded. The imported model registries contain all fifteen rank-four and all seventeen rank-five realized based-ring types, checked up to multiplicities 128 and 20, respectively; all encountered subrings fall within these ranges. The complete rank-six multiplicity-two classification [17] supplies context and methodology, without being used as an unstated rank-seven classification theorem.

Proposition 2 (The formal-codegree test at rank seven)**.** *Suppose that R is a rank-seven fusion ring, possibly noncommutative. Let $C = \sum_{i=0}^{6} N_i N_i^{\mathsf{T}}$ be its regular Casimir matrix. If an irreducible factor of the characteristic polynomial of C is not the minimal polynomial of a d-number, then R cannot be the Grothendieck ring of a complex fusion category.*

Proof. In the commutative case, the eigenvalues of C are the formal codegrees of one-dimensional characters and must be d-numbers by Ostrik [24]. In the noncommutative case, semisimplicity, $\dim_{\mathbb{C}}(R \otimes \mathbb{C}) = 7$, and the positive one-dimensional character imply the unique possible decomposition $R \otimes \mathbb{C} \simeq \mathbb{C}^3 \oplus M_2(\mathbb{C})$. There is only one two-dimensional irreducible module; it is fixed by every field automorphism, so its formal codegree is a rational algebraic integer and hence an integer. The Casimir acts on this matrix block by twice that integer. Every remaining nonrational

Casimir eigenvalue comes from a one-dimensional character and is a formal codegree, so Ostrik's condition applies. The implementation factors the exact integral characteristic polynomial and tests the coefficient-divisibility characterization of d-numbers. The replay checker recomputes and factors that polynomial. □

If S is a proper commutative fusion subring of a hypothetical categorification, the quotient $\mathrm{FPdim}(R)/\mathrm{FPdim}(S)$ must be an algebraic integer [20]. The subcategory-divisibility stage computes this quotient in the exact Frobenius–Perron number field, records its characteristic polynomial, and certifies a nonintegral coefficient. Its certificate is rederived from the complete tensor.

3 Complete oriented tests at rank seven

The oriented-dimension-order row of Table 2 uses the general, *oriented* signed-sphericalization construction established in [17]. For an arbitrary hypothetical category $\mathcal{C}$, its canonical sphericalization has two lifts of each simple object, interchanged by a sign object ε. Taking the quotient by $(\varepsilon+1)$ produces integer coefficients T_{ij}^k with $|T_{ij}^k| \le N_{ij}^k$ and $T_{ij}^k \equiv N_{ij}^k \pmod 2$. Unit, duality, cyclic Frobenius reciprocity, and ordered associativity further constrain the quotient. The self-dual positive-indicator normalization imposes the established square-channel congruences modulo four. These conditions are necessary and cover every categorification, whether pivotal or not; a signed solution is not itself a categorification.

The rank-six proof of the general construction and the dimension-order obstruction [17] are rank-independent. Here their enumeration engine is extended to seven basis elements. It enumerates the independent cyclic/dual-reflection coefficient orbits, prunes only exact associativity equations with checked integer bounds, and constructs *every* signed quotient. For each quotient the checker verifies either its adjoint parity, its gauge equivalence with the original ring, or the failure of every admissible spherical character by codegree and dimension-order tests. In particular the dimension-order test requires the central-induction dimension δ/f_χ to lie in the $\mathbb{Z}$-span of the signed simple dimensions, as required by [25, 17].

Proposition 3 (Verified exhaustive branches)**.** *The oriented-dimension-order criterion excludes four, fifty-one, sixty-one, and one hundred thirty previously surviving rank-seven rings of exact multiplicities one through four, respectively. For* R7_1_5*,* R7_1_23*,* R7_1_33*, and* R7_1_38*, the complete signed enumerations have respectively* $2, 4, 4, 8$ *associative quotient tables. Every table in each of these 246 certified exhaustive searches is eliminated by necessary conditions for arbitrary complex fusion categories.*

Proof. The machine-readable proof tree for each of the 246 rings gives its full dimension-character factorization, every obstruction, the ordered signed-orbit domain, and one checked branch for every enumerated table. The independent verifier rebuilds the domains from the source tensor, repeats the bounded exhaustive enumeration, checks the *entire* list of surviving tables for equality to the recorded list, and recomputes each branch witness. The search used to discover these finite trees is logically separate from their exhaustive replay; timed-out or unfinished searches generate no negative decision. The enumeration returns the four displayed multiplicity-one branch counts and no uncovered branch for any of the 246 certificates. The rank-independent necessity of the search is the sphericalization theorem cited above. □

4 Further theoretical exclusions

The local triangular-prism proof of Liu–Palcoux–Ren–Vercleyen [6, Corollary 6.2, Theorem 6.4 and Theorem 7.2] excludes $R7_2_185$ and $R7_3_635$ over $\mathbb{C}$. The first is exactly their $\mathcal{F}_{210}$. The second is a consequence of the same local proof, as follows. In the source basis of dimensions $(1, 5, 5, 5, 6, 7, 7)$, their tensors agree outside the cube on the two dimension-seven objects. Both have this strictly positive integral dimension character, so any categorification is pseudo-unitary and has the canonical spherical structure. The cited proof uses the squares of two dimension-five objects k, l, selected columns of dimensions one and five, their dualities and categorical dimensions. The zero constraints are read from the squares of those selected objects. Thus none of the differing coefficients on the dimension-seven cube enters either local subsystem or its linking equation. Both rings give precisely the inconsistent polynomial system in the proof of [6, Theorem 7.2]. The supplementary checker verifies all these tensor equalities and the positive dimension character; it does not rerun the published Gröbner-basis calculation.

For $R7_3_900$ and $R7_3_991$, a different short argument gives a general exclusion. Their dimension type is $(1, 1, 1, 1, 2, 4, 4)$ and their Frobenius–Perron dimension is $40 = 2^3 \cdot 5$. Any categorification would be solvable [7, Theorem 1.6], hence weakly group-theoretical and unitarizable [8, Theorem 5.20]. For a unitary categorification, the positive pth-root limit of [9, Corollary 3.4] requires, for every subset S of basis labels and $N_{ij}^l > 0$,

$$\frac{\sum_{x\in S} N_{lk}^x d_x}{d_l} \le \frac{\sum_{m\in T} N_{jk}^m d_m}{d_j}, \qquad T = \{m : N_{jk}^m > 0,\ N_{im}^x > 0 \text{ for some } x \in S\}.$$

Taking S to be the objects of dimension at most two, the exact witnesses in the supplement give $2 \le 1$ for each ring. The first witness is $(i, j, k, l) = (1, 1, 2, 1)$; the second is $(3, 4, 4, 3)$, using zero-based census labels. This unitary obstruction is a general one here because the dimension theorem forces unitarizability.

Proposition 4 (A half-partition criterion)**.** *Let R be an integral commutative fusion ring of dimension $4pq$, for distinct odd primes p, q. Write $a_E = 4pq/f_E$ for its formal codegrees. Suppose an odd-prime-power $a_E > 1$ occurs, and every submultiset of the a_E excluding the unit and summing to $2pq$ leaves an odd-prime-power entry in its complement. Then any categorification is weakly group-theoretical and has the Frobenius property.*

Proof. Integrality gives the canonical positive spherical structure. By [11, Theorem 2.13], the induced unit $I(1)$ in the center has distinct multiplicity-one summands of dimensions a_E. An odd-prime-power summand implies a nontrivial symmetric subcategory of the center [7, Corollary 7.2]. If this provides no nontrivial Tannakian subcategory, it must be sVec. For its fermion f, [11, Proposition 2.8] gives $s[I(1)] = 4pq[I(1)]$. The fermion has twist -1, so it is absent from $I(1)$, whose twist is trivial. Its row therefore gives $\sum_E \epsilon_E a_E = 0$, with monodromy signs $\epsilon_E = \pm 1$ and $\epsilon_{\mathbf{1}} = 1$. The negative signs form a half-partition. By hypothesis an odd-prime-power summand centralizes f. Its slightly degenerate centralizer consequently contains a nontrivial Tannakian subcategory by [7, Proposition 7.4].

Write this subcategory as $\mathrm{Rep}(G)$. Its de-equivariantization has neutral dimension $(4pq/|G|)^2$, hence $|G|$ divides $4pq$. If p or q divides $|G|$, the neutral category has at most two prime divisors and is solvable. Extension and equivariantization reconstruct the center; Morita invariance and subcategory closure then make the original category weakly group-theoretical [7, Theorem 1.6 and Proposition 4.1]. Otherwise $|G|$ is two or four and supplies $\mathrm{Rep}(C_2)$. The original category is a C_2-extension or C_2-equivariantization of a category of dimension $2pq$ [7, Propositions 2.9–2.10].

That category is integral: in the equivariantization case its simple dimensions are rational quotients of integral dimensions by orbit sizes and projective representation degrees, hence integers. The integral squarefree-dimension theorem [7, Theorem 9.2] makes it group-theoretical. The Frobenius property follows from [7, Theorem 1.5]. □

For $R7_3_905$, the exact positive dimension vector is $(1, 6, 6, 1, 1, 1, 8)$, and the formal codegrees are $(140, 20, 20, 7, 4, 4, 4)$. Thus the induced-unit dimensions are $(1, 7, 7, 20, 35, 35, 35)$. The only half-partition excluding the unit consists of two dimension-35 summands, leaving both dimension-seven summands. The proposition applies, but 6 does not divide 140. This excludes every complex categorification. The checker verifies the entire formal-codegree polynomial and all 64 nonunit subsets exactly.

Finally, $R7_4_2727$ has dimension type $(1, 1, 1, 3, 6, 6, 6)$, dimension 120, and formal codegrees $(120, 60, 60, 24, 4, 3, 3)$. The induced unit has simple summands of dimension two, so the center contains a nontrivial symmetric subcategory. The invertible group has odd order three; [12, Lemma 2.5] therefore gives a nontrivial Tannakian subcategory, and [12, Lemma 3.1] makes any categorification weakly group-theoretical and hence unitarizable. The exact support inequality above gives $4 \leq 2$ for $(i, j, k, l) = (3, 4, 3, 3)$ and $S = \{6\}$, a contradiction.

4.1 A double-equivariantization obstruction in dimension 52

The ring $R7_2_75$ has dimension type $(1^4, 4^3)$, invertible group V_4, and trivial universal grading. Every invertible fixes every dimension-four simple. These are exact tensor properties checked by the supplement. Suppose $\mathcal{C}$ categorifies it. Dimension $52 = 4 \cdot 13$ makes $\mathcal{C}$ solvable [7, Theorem 1.6]. The last step of its prime cyclic extension/equivariantization construction [7, Proposition 4.4] cannot be an extension, since $\mathcal{C}_{\mathrm{ad}} = \mathcal{C}$. Its character group must embed into V_4, so the last step is $\mathcal{C} = \mathcal{D}^{C_2}$, with $\mathrm{FPdim}(\mathcal{D}) = 26$. Each possible nontrivial character fixes the three dimension-four simples and pairs the four invertibles. The equivariant-simple orbit/stabilizer formula [1, Proposition 4.26] therefore forces $\mathcal{D}$ to have type $(1^2, 2^6)$ and rank eight. A fixed simple of the action has two lifts exchanged by the character; a free orbit has one lift of twice the dimension. There are no higher-dimensional projective representations of C_2.

The category $\mathcal{D}$ also has trivial universal grading. Equal dimensions of faithful grading components [20, Proposition 8.20] make the grading-group order divide 26. Orders 13 and 26 exceed the rank. Order two would give a neutral category of prime dimension 13, necessarily pointed and of rank 13 [20, Corollary 8.30], also impossible. Applying the same solvable last-step argument gives

$$\mathcal{D} = \mathcal{P}^{C_2}, \qquad \mathcal{P} = \mathrm{Vec}^{\omega}_{C_{13}}.$$

The inner C_2 must act nontrivially on C_{13}, hence by inversion; a trivial object action would give 26 invertible simples in $\mathcal{D}$.

The canonical central $\mathrm{Rep}(C_2)$ in $\mathcal{D}$ is unique over its underlying two invertibles. Indeed, the kernel of $\mathrm{Inv}(Z(\mathcal{D})) \to \mathrm{Inv}(\mathcal{D})$ is the group of tensor automorphisms of $\mathrm{Id}_{\mathcal{D}}$. Such an automorphism is trivial on every summand of $Y \otimes Y^*$, hence on $\mathcal{D}_{\mathrm{ad}} = \mathcal{D}$. Two lifts of the same invertible differ by this kernel. The outer C_2 action thus preserves the central $\mathrm{Rep}(C_2)$. Equivariantizing it gives a central Tannakian $\mathrm{Rep}(G)$ with fully faithful underlying embedding into $\mathcal{C}$, where $1 \to C_2 \to G \to C_2 \to 1$. Its symmetric fibre functor is the composite of the two forgetful functors. Therefore G is C_4 or V_4; its four characters exhaust the invertibles of $\mathcal{C}$, forcing $G = V_4$. De-equivariantization in stages now gives $\mathcal{C} = (\mathrm{Vec}^{\omega}_{C_{13}})^{V_4}$, with the normal C_2 acting by inversion

[7, Section 2.6 and Proposition 2.10] [1, Theorem 4.18(i) and Proposition 4.19]. These are central Tannakian constructions; no braiding of $\mathcal{C}$ is assumed.

The object action $V_4 \to \mathrm{Aut}(C_{13}) = C_{12}$ has image of exponent at most two and contains inversion. Every nonidentity object consequently has orbit size two and stabilizer C_2. Each of the six nonidentity orbits contributes two dimension-two simples, while the unit contributes the four characters of V_4. This forces type $(1^4, 2^{12})$, contradicting $(1^4, 4^3)$. Thus $R7_2_75$ has no complex fusion categorification, without any pivotal or unitary assumption. The proof does not require ω to be trivial.

4.2 An order-sixteen orbit obstruction in dimension 96

The same method excludes $R7_2_252$ and $R7_2_315$. Both have dimension type $(1, 1, 2, 3, 3, 6, 6)$, invertible group C_2, and trivial universal grading. Their nontrivial invertible h fixes the dimension-two and dimension-six simples and exchanges the two dimension-three simples. Solvability and the last-step argument force $\mathcal{C} = \mathcal{D}^{C_2}$. The orbit/stabilizer formula then gives $\mathcal{D}$ dimension 48 and type $(1^3, 3^5)$.

The universal grading of $\mathcal{D}$ is trivial. Otherwise its group has order dividing 48, is solvable by Burnside's theorem, and has a prime cyclic quotient of order two or three. Equal component dimensions [20, Proposition 8.20] make both impossible. For a C_2 grading all three invertibles lie in the neutral component, requiring $24 = 3 + 9n$. For a C_3 grading their group maps trivially or isomorphically to the grading group, requiring $16 = 3 + 9n$ or $16 = 1 + 9n$. Thus solvability forces $\mathcal{D} = \mathcal{E}^{C_3}$ with $\mathrm{FPdim}(\mathcal{E}) = 16$. Its unit accounts for all three invertibles of $\mathcal{D}$. A dimension-three simple of $\mathcal{D}$ comes from either a free orbit of three invertibles of $\mathcal{E}$ or a fixed dimension-three simple, the latter giving three such simples in $\mathcal{D}$. Counting the five dimension-three simples leaves only types (1^{16}) or $(1^7, 3)$ for $\mathcal{E}$. The second is impossible: its seven invertibles would fix its unique noninvertible Y, so YY^* would contain all seven, leaving dimension two that cannot be filled by copies of Y. Consequently $\mathcal{E} = \mathrm{Vec}_H^\omega$ for a group H of order sixteen.

As in the preceding proof, $\mathcal{D}_{\mathrm{ad}} = \mathcal{D}$ makes the canonical central $\mathrm{Rep}(C_3)$ unique over its underlying invertibles. The outer C_2 action preserves it. Equivariantization in stages gives $\mathcal{C} = (\mathrm{Vec}_H^\omega)^G$, where $1 \to C_3 \to G \to C_2 \to 1$ and the central $\mathrm{Rep}(G)$ embeds fully faithfully into $\mathcal{C}$. The cyclic group of order six would give six invertibles; hence $G = S_3$. The identity orbit contributes dimensions $1, 1, 2$, exhausting the small simples. Any other fixed orbit has projective stabilizer S_3, whose irreducible projective degrees are at most $\sqrt{6}$, and would give an extra small simple. An orbit of size two has stabilizer C_3 and would give extra dimension-two simples. An orbit of size three gives two dimension-three simples, and a free orbit gives one dimension-six simple. Thus the action on H must have orbit sizes

$$(1, 3, 6, 6).$$

Its normal C_3 acts without fixed points outside the identity; a transposition fixes exactly two elements of H.

Here is a purely group-theoretic contradiction. For a fixed-point-free order-three automorphism α of a group of order sixteen, every characteristic subgroup has order congruent to one modulo three. Thus the center has order four or sixteen. If it has order four, it is V_4, and the noncyclic quotient $H/Z(H)$ is also V_4. The commutator is an α-equivariant alternating map $\bigwedge^2 V_4 \to Z(H)$. Its source has trivial C_3 action, whereas its target has no nonzero fixed vector. It vanishes, a contradiction. So H is abelian. The characteristic subgroups $2H$ and $4H$ exclude C_{16}, $C_8 \times C_2$, and $C_4 \times C_2^2$, leaving $H = C_4^2$ or C_2^4.

Every involution of either remaining group fixes at least four elements. For C_2^4, $(S-I)^2=0$ gives $\dim\ker(S-I)\geq 2$. For C_4^2, if S reduces to the identity modulo two, it fixes all four elements of $2H$. Otherwise a lifted basis change gives $S=\left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right)$ with a,b,d odd and c even. From $S^2=I$ one obtains $bc=0$ and $b(a+d)=0$, hence $c=0$ and $d=-a$ modulo four. Each of the four choices of x has the unique fixed-vector completion $y=-b^{-1}(a-1)x$; the second equation follows from $a^2=1$. This contradicts the required two fixed points and excludes both rings over $\mathbb{C}$. The supplement checks the exact tensor premises and also provides an independent finite corroboration of this theoretical lemma.

5 Exact constructions and replay

The three formerly undecided multiplicity-one rings match AnyonWiki $\mathrm{FR}_3^{7,2}$, $\mathrm{FR}_{12}^{7,2}$, and $\mathrm{FR}_3^{7,4}$, respectively, by full based-ring isomorphisms. For each, we use the published category with code $(7,1,d,i,1,0,1)$ in the Vercleyen–Slingerland census [13]. The symbolic associators supplied through TensorCategories.jl lie over $\mathbb{Q}(\zeta_8)$, $\mathbb{Q}(\zeta_{28})$, and $\mathbb{Q}(\zeta_{16})$, respectively. Independent rational-polynomial replay checks all 9,856, 45,061, and 9,856 admissible pentagon equations, all 585, 837, and 585 nonempty associator blocks for invertibility, and the unit and rigidity constraints. These checks establish complex fusion categories directly, with no pivotality assumption. The AnyonWiki category list additionally records unitary gauges. Only existence of a model is used: neither completeness nor uniqueness of the external category census is required. The supplementary evidence records the exact category codes, cyclotomic coefficients, source hashes and basis permutations.

The positive model catalogue constructs complete fusion tensors from the cyclic group, ordinary finite-group character products, the Tambara–Yamagami rule, $SU(2)_k$ truncation, its even parts, odd metaplectic rules, and the categorical $\mathbb{Z}/2$ zest of $SU(2)_6$ [27, 21, 23, 28, 29]. Character products for $\mathrm{Rep}(S_5)$, $\mathrm{Rep}(SL_2(\mathbb{F}_3))$, $\mathrm{Rep}(C_{13}\rtimes C_3)$ and $\mathrm{Rep}(C_{11}\rtimes C_5)$ are computed with exact characters in the bundled software. Two further complete model tensors come from the published noncommutative integral appendix of [18, Theorem 1.6 and Section 6.2]: the representation category of its dimension-24 Kac algebra and the group-theoretical category $\mathcal{C}(A_5,1,A_4,1)$. The published full tensors and their byte digest are pinned in the package. For each constructed or published model, the program validates the fusion axioms, then checks all 6! unit-preserving basis permutations against every compatible census tensor. A positive record carries its explicit permutation. Different categorical structures may share a based ring; the count is of rings, not categories.

Three additional standard constructions give full-tensor matches: $R7_3_602 = K_0(\mathrm{Rep}(A_6))$, $R7_2_218 = K_0(\mathrm{Rep}(C_{13}\rtimes_5 C_4))$, and $R7_2_313 = K_0(\mathrm{Adj}(\mathrm{SU}(3)_5))$. The first uses exact character products over $\mathbb{Q}(\sqrt{5})$; the second uses induction of characters of C_{13}, with the complement generator acting by exponent five. The third is the triality-zero subcategory of the standard unitary quantum-group category [10]; exact Weyl alternants over $\mathbb{Q}(\zeta_{24})$ reconstruct all its fusion coefficients. Each is checked at all 343 structure constants.

5.1 Two dimension-24 equivariantizations

Put $\mathcal{T}=\mathrm{TY}(C_3,\chi,1/\sqrt{3})$, with $\chi(a,b)=\exp(2\pi iab/3)$. Write m for its noninvertible simple and $|a|=0$, $|m|=1$ for its C_2 grading. Inversion $I(a)=-a$, $I(m)=m$ preserves χ and gives a coherent unitary C_2 action with canonical channel identifications as tensorators and identity square coherence [4, Section 4A and Proposition 4.8]. The associators are phases, permutations or

the normalized Fourier matrix.

First take $\mathcal{E} = \mathcal{T} \boxtimes \mathrm{Vec}(C_2)$ and $F(X, e) = (I(X), e + |X|)$. Grading additivity and the trivial pointed associator give the tensor structure, and $F^2 = \mathrm{Id}$ as a tensor functor. Thus $\mathcal{E}^{C_2}$ is unitary. Its two fixed simple objects $(0, e)$ each have two equivariant lifts t_e^+, t_e^-; the other invertibles form orbits $P_e = \{(1, e), (2, e)\}$, and the two odd objects form $Z = \{(m, 0), (m, 1)\}$. In the ordered basis

$$(1, h, t, ht, P_0, P_1, Z) = (t_0^+, t_0^-, t_1^+, t_1^-, P_0, P_1, Z),$$

the invertible group is C_2^2, h fixes both P_e, t exchanges them, and all invertibles fix Z. The remaining products are

$$\begin{aligned} P_e P_f &= t_{e+f}^+ + t_{e+f}^- + P_{e+f}, & P_e Z &= Z P_e = 2Z, \\ Z^2 &= 1 + h + t + ht + 2P_0 + 2P_1. \end{aligned}$$

Here and below subscripts are modulo two. Fixed output channels are exchanged in pairs and have one invariant vector of each sign; free orbit outputs have a unique equivariant lift. These rules agree with $R7_2_1$ under the model-to-census permutation $(0, 5, 1, 2, 4, 3, 6)$.

For the second construction, let z generate C_4 and use the normalized unitary cocycle

$$\omega(a, b, c) = i^{w(a,b,c)}, \qquad w(a, b, c) = a \left\lfloor \frac{b + c}{4} \right\rfloor, \quad 0 \le a, b, c < 4.$$

Inside $\mathcal{T} \boxtimes \mathrm{Vec}(C_4, \omega)$ take the full diagonal subcategory with simples (a, z^{2e}), (m, z) and (m, z^3). It is closed under tensor products and duality because TY parity equals the pointed degree modulo two. Define F by inversion in both factors. Its TY tensorator is that of I; on pointed degrees use $J(a, b) = i^{u(a,b)}$, in the direction $F(X) \otimes F(Y) \to F(X \otimes Y)$, where

$$u = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 3 & 2 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Reducing arguments modulo four, direct substitution gives

$$\begin{aligned} u(b, c) - u(a + b, c) + u(a, b + c) - u(a, b) &\equiv w(a, b, c) - w(-a, -b, -c), \\ u(a, b) + u(-a, -b) &\equiv 0 \pmod 4. \end{aligned}$$

The first is tensor-functor coherence; the second makes F^2 the identity tensor functor. Identity multiplication coherence therefore defines a genuine C_2 action. The supplement checks all 256 pointed cocycle equations, 64 tensorator equations and 16 square equations.

The equivariantization has the same simple-orbit pattern. The positive lift t of $(0, z^2)$ satisfies $t^2 = h$, since $J(2, 2) = -1$; hence its invertibles form C_4. All other displayed products remain valid, with $t_1^+ = t$ and $t_1^- = t^3$. In Z^2, the degree-two pair of channels is exchanged with coefficients $1, 1$, and the degree-zero pair with coefficients $i, -i$. Both involutions have trace zero and eigenvalues $1, -1$, giving one copy of each fixed lift. The complete tensor matches $R7_2_195$ under $(0, 6, 1, 2, 5, 3, 4)$. All scalars and tensorators are unitary, so both constructions are unitary categories over $\mathbb{C}$. Besides comparing all 686 coefficients, the replay independently reconstructs them as C_2-invariant dimensions of equivariant Hom spaces, using $(\dim + \mathrm{tr}(F))/2$.

5.2 A quaternion equivariantization in dimension 48

Let $Q_8 = \{\pm 1, \pm e_1, \pm e_2, \pm e_3\}$ with $e_1e_2 = e_3$, $e_2e_3 = e_1$, $e_3e_1 = e_2$, and $e_a^2 = -1$. For $p \in S_3$, the rule

$$\rho_p(e_a) = \operatorname{sgn}(p) e_{p(a)}$$

is a group automorphism, and $\rho_p\rho_q = \rho_{pq}$. It gives a strict unitary action on $\mathrm{Vec}(Q_8)$, so $\mathrm{Vec}(Q_8)^{S_3}$ is a unitary fusion category of dimension 48. Each fixed central grade $1, -1$ carries the three ordinary irreducibles $1, \varepsilon, V$ of S_3; the six noncentral grades form one free orbit, giving a simple X of dimension six [5]. Label the first six simples (z, a), with $z \in C_2$ and $a \in \{1, \varepsilon, V\}$. Their products are those of $\mathrm{Rep}(S_3) \boxtimes \mathrm{Vec}(C_2)$ and $(z, a)X = X(z, a) = \dim(a)X$. In X^2, the six ordered pairs with product either central grade form a regular S_3-set; each noncentral grade has four ordered pairs. Consequently, in the order $(0,1), (0,\varepsilon), (0,V), (1,1), (1,\varepsilon), (1,V), X$,

$$X^2 = (0,1) + (0,\varepsilon) + 2(0,V) + (1,1) + (1,\varepsilon) + 2(1,V) + 4X.$$

This complete tensor matches $R7_4_2$ under $(0, 5, 4, 1, 2, 3, 6)$. The replay checks the group action, computes the central permutation characters and compares all 343 coefficients.

5.3 The twisted quaternion model from a Chern class

The remaining dimension-48 ring $R7_4_2019$ has a unitary group-theoretical model

$$\mathcal{C}(\Gamma, \Omega, S_3, 1), \qquad \Gamma = Q_8 \rtimes S_3, \qquad [\Omega] = c_2(V) - \inf c_2(R).$$

The action is the signed-axis action above. To construct the class, view unit quaternions inside $\mathrm{SU}(2)$, and distinguish the scalar imaginary unit I from quaternion units i, j, k. Set

$$r = -\frac{1 + i + j + k}{2}, \qquad s = I\frac{i - j}{\sqrt{2}}.$$

Then $r^3 = s^2 = 1$ and $srs = r^{-1}$; conjugation implements the signed axis cycle and transposition. Thus r, s define a unitary representation $R : S_3 \to U(2)$ with determinant the sign character, and $V(q, p) = qR(p)$ is a representation of Γ. The ordinary second Chern classes give a well-defined class in $H^4(B\Gamma, \mathbb{Z}) \cong H^3(\Gamma, U(1))$ [3]. Its restriction to S_3 is zero.

Put $z = -1 \in Q_8$ and $K = \langle z \rangle \times S_3$. If $x = c_1(L)$ for the sign line of $\langle z \rangle$ and $y = c_1(\det R)$, the splitting principle gives

$$[\Omega]|_K = c_2(L \otimes R) - c_2(R) = x^2 + xy.$$

Writing elements of K as (a, p) and parity as $\epsilon(p)$, this restriction has the explicit normalized representative

$$\omega_K((a,p),(b,q),(c,r)) = (-1)^{abc + a\epsilon(q)\epsilon(r)}.$$

Indeed the real logarithm $a(bc + \epsilon(q)\epsilon(r))/2$ has integer coboundary $ab(cd + \epsilon(r)\epsilon(s))$, representing $x^2 + xy$. The replay checks all 12^4 such identities exactly. Choose a unitary global cocycle representing $[\Omega]$. Extend a correcting normalized two-cochain from K to Γ to make its restriction exactly ω_K. Then $\Omega|_{S_3} = 1$. The group-theoretical category is the category of $\mathbb{C}[S_3]$-bimodules in $\mathrm{Vec}_\Gamma^\Omega$ [2, Section 3 and Theorem 5.1]. The unitary cocycle and standard positive separable subgroup algebra make it unitary. This proves existence without constructing a table of 48^3 cocycle values.

The central double cosets have stabilizer S_3. Over z, the multiplier is

$$\alpha_z(p,q) = \frac{\omega_K(z,p,q)\omega_K(p,q,z)}{\omega_K(p,z,q)} = (-1)^{\epsilon(p)\epsilon(q)}.$$

It has the one-dimensional projective representation $u(p) = I^{\epsilon(p)}$. Its lift t satisfies $t^2 = \varepsilon$: the central-degree tensorator is trivial and $u(p)^2$ is the sign character. Thus the six central simples are $1, \varepsilon, v, t, t\varepsilon, tv$, with the S_3 representation products and relation $t^2 = \varepsilon$. The noncentral double coset is free and gives one simple X of dimension six. Products $YX = XY = \mathrm{FPdim}(Y)X$ follow from support and dimension. Each central product fibre of X^2 is a free transitive S_3-set, so its projective monomial representation is projective regular, with irreducible multiplicities $1, 1, 2$ regardless of the cocycle phases. The noncentral fibres each have size four. Hence

$$X^2 = 1 + \varepsilon + 2v + t + t\varepsilon + 2tv + 4X.$$

In this order the model-to-census permutation is $(0, 6, 5, 1, 2, 3, 4)$. The supplement checks the representation relations, action, characteristic-class restriction, projective lift, and all 343 coefficients. The remaining 59 tensors are undecided.

The complete package contains the pinned gzip source, analysis code, negative proof trees, model tensors, the stage-count summary and a second entry point, `verify.py`. Run `python analyze.py -max-m 4 -dnumber` and then `python verify.py`. The verifier rechecks the source digest and every row, every full-tensor model isomorphism, each stage total and each exclusion certificate, including the exhaustive signed lists. Running `python -O verify.py` gives the same checks without relying on Python's optional `assert` statements. For the dated update alone, `python software/verify_residual_20261002.py` (also with `-O`) checks the seven new models and nine new exclusions, including the coherent actions, the Chern-class construction, and their fusion rules. It verifies preservation hashes for all 4185 unaffected records and all 4107 previous exclusions; the unchanged costly searches are not repeated by this shorter entry point.

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